



Pharmacology 1 (PYCH304)

Drugs Used in Osteoarthritis & Rheumatoid Arthritis

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Objectives



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After this class, students should be able to

- understand the pathophysiology of Osteoarthritis (OA) and Rheumatoid Arthritis (RA) and related therapeutic targets
- **identify the major classes of drugs** used to treat OA and RA.
- **explain the mechanisms of action** of key drugs used in OA and RA.
- **know the side effects** and **contraindications** of commonly prescribed drugs for OA and RA.



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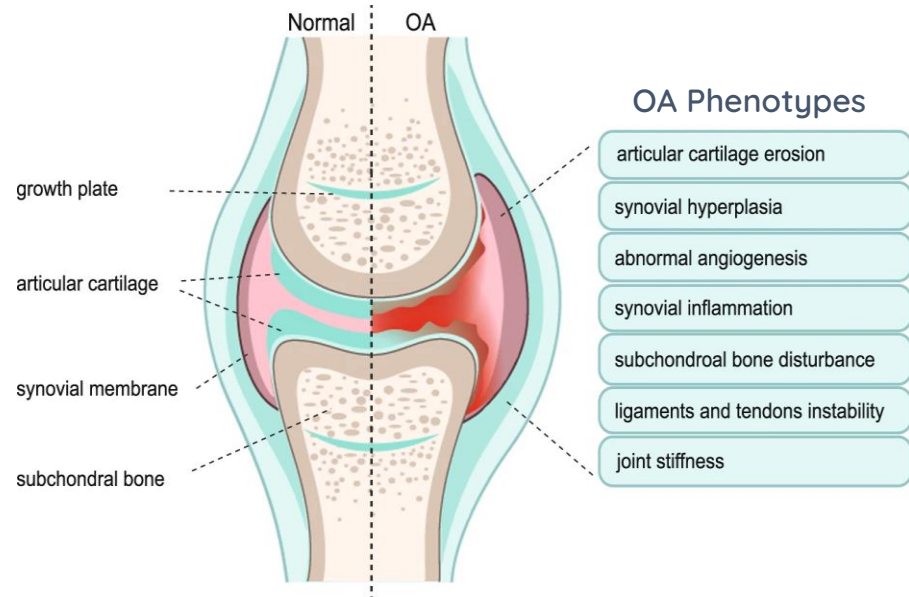
Osteoarthritis

โรคข้อเสื่อม/ข้อกระดูกอ่อนเสื่อม



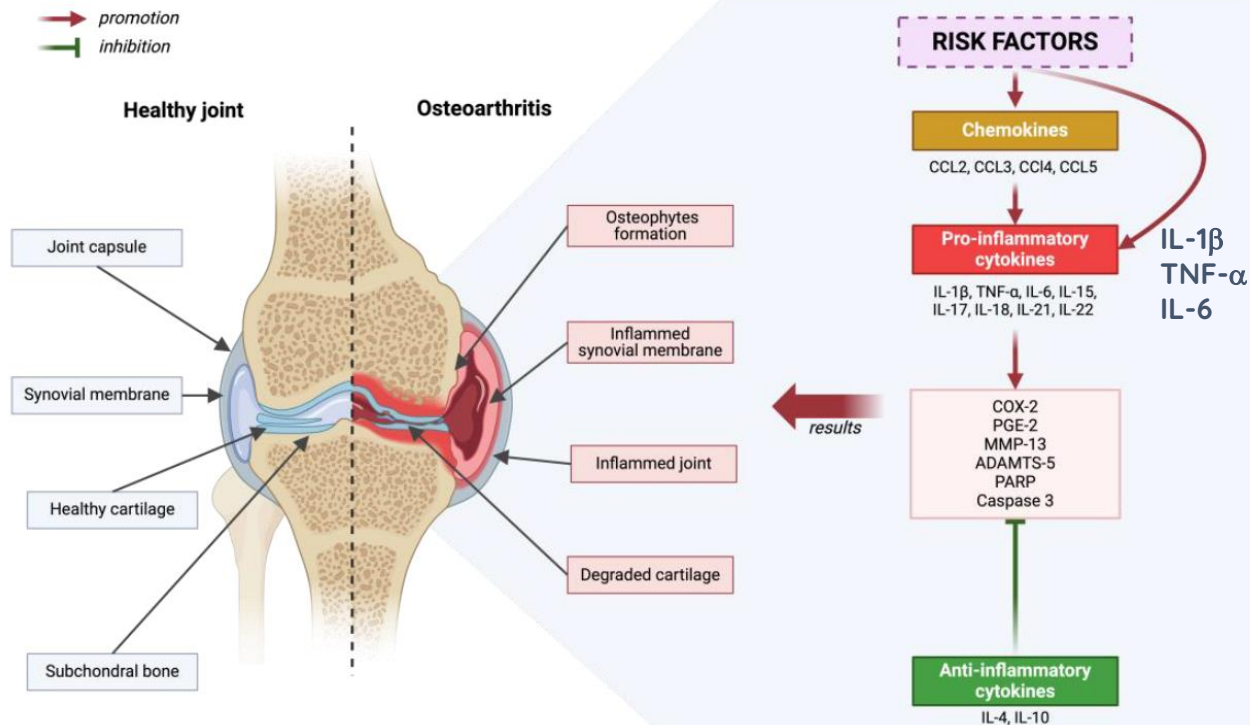
Osteoarthritis (OA)

- Definition: a **chronic degenerative joint disorder** associated with cartilage destruction, subchondral bone remodeling & synovial membrane inflammation.
- is a **slow progressive** disease.
- is the **most common type of arthritis**. (e.g., hand, knee, hip)
- high prevalence in the **elderly**.
- is a **leading cause of disability** among older adults.





Pathophysiology of OA



Anabolic ↓
Catabolic ↑
Inflammatory responses ↑



Goals:

OA Treatment

ACR / EULAR / OARSI

- Reduce pain and swelling.
- Improve joint function and minimize disability.
- Improve the patient's quality of life.

Non-pharmacological

- Patient education
- Physical therapy
- Exercise
- Weight loss
- Heat & cold therapy
- Assistive device
- Surgery

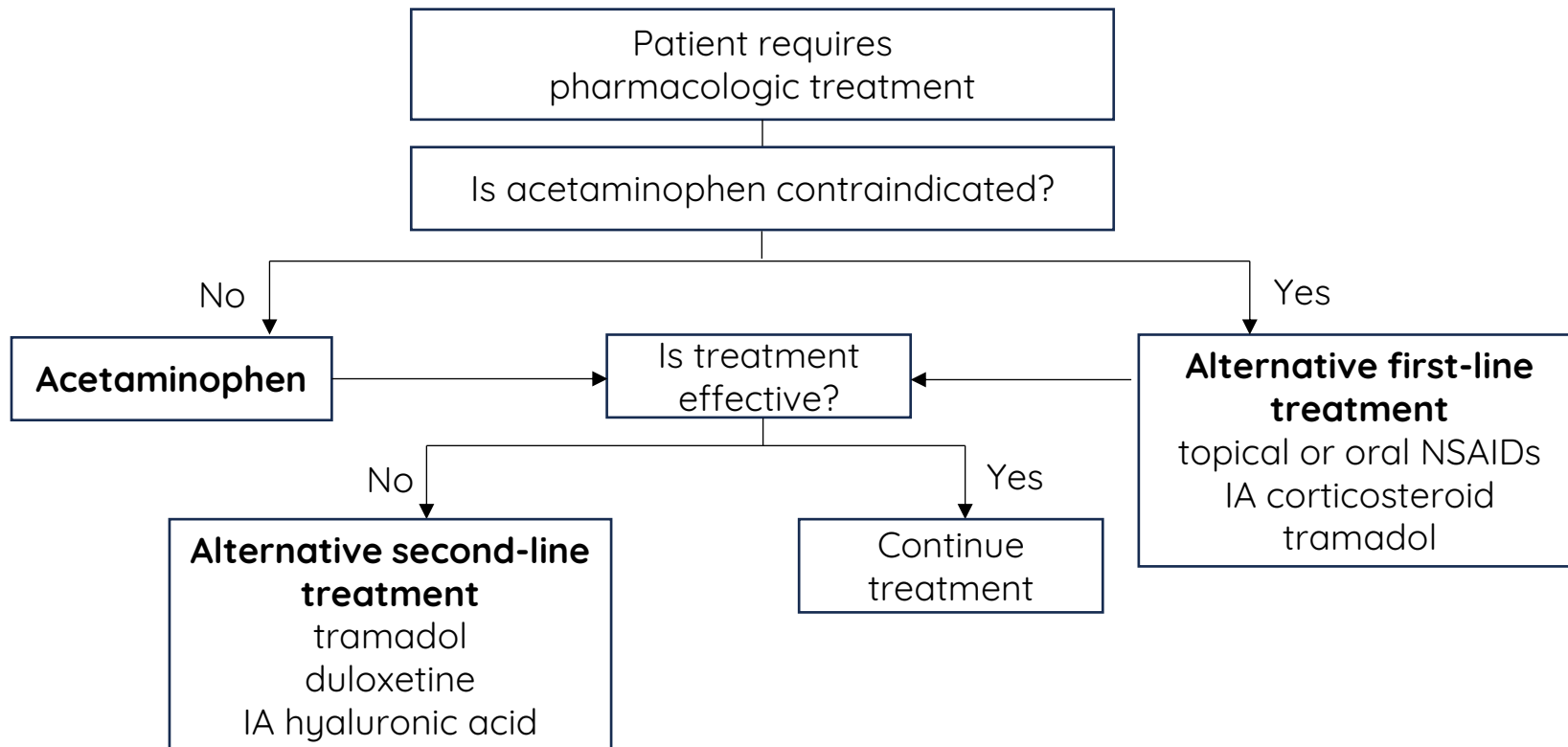
Pharmacological

- | | |
|----------------------|---------------|
| • Acetaminophen | • Glucosamine |
| • NSAIDs | • Chondroitin |
| • Tramadol | • DMOADS |
| • Duloxetine | |
| • IA corticosteroid | |
| • IA hyaluronic acid | |

Pharmacological treatment



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Acetaminophen

- **First-line for mild to moderate pain.**
- Good risk-benefit profile.
- No anti-inflammatory activity.
- **Doses**
 - 500 – 1000 mg, 3-4 times/day (maximum single dose 1000 mg)
 - Maximum dose:
 - Adult: 4000 mg/day (some guidelines: 3000 mg/day to reduce liver toxicity risk)
 - Children: 50-75 mg/kg/day in children.
 - Elderly patients/patients with liver disease: 2000-3000 mg/day



Acetaminophen

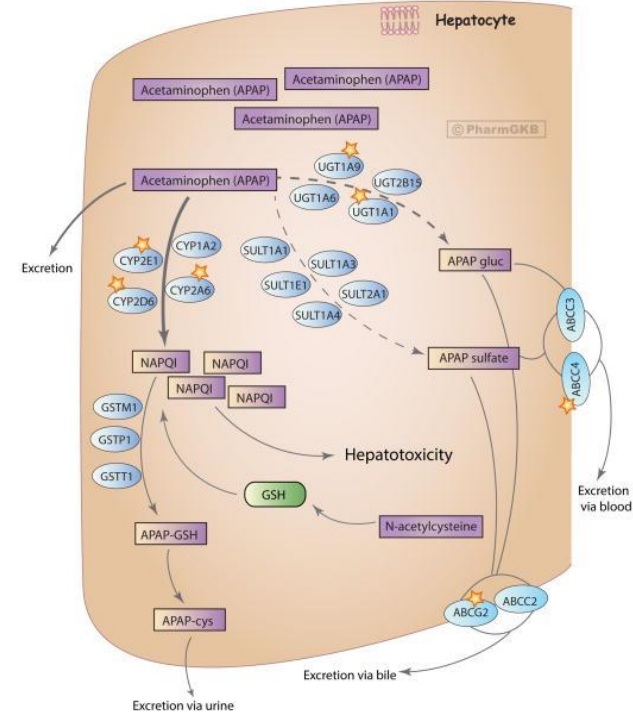
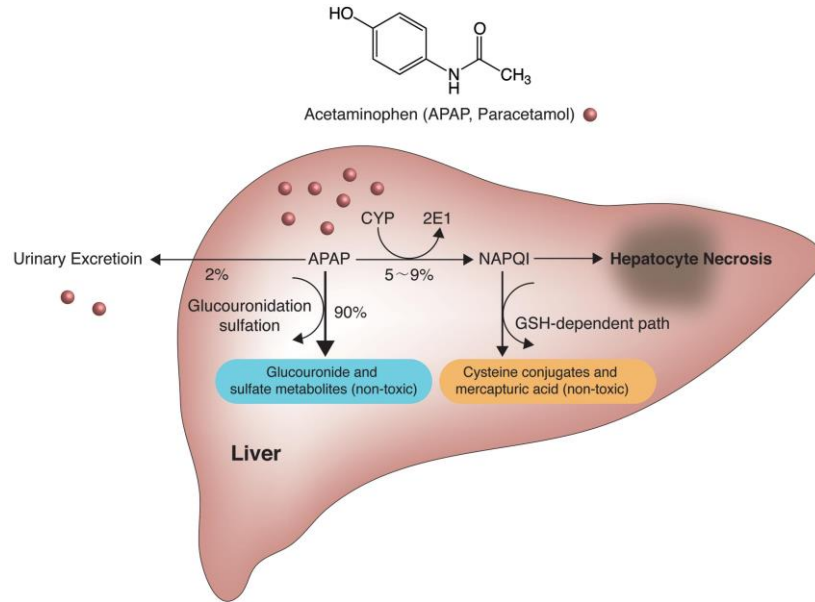
- **Pharmacokinetics**

- Oral bioavailability: 88%
- Peak plasma concentration: 1.5 hours
- Plasma half-life: 1.5-2.5 hours.
 - Overdose/Impaired metabolism: prolonged half-life → 4-8 hours
- Metabolism:
 - mainly liver: CYP2E1 (main), CYP1A2, CYP3A4
 - kidney
 - Intestine
- Excretion: most metabolites are excreted via urine. (90% of the administered dose is eliminated within 24 hours.)

Acetaminophen metabolism & liver hepatotoxicity



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Factors influencing acetaminophen-related hepatotoxicity

Factors	Clinical implications
Dose and pattern of use	↑ APAP toxicity with acute, high doses
EtOH	↓ APAP toxicity with acute EtOH ingestion
	↑ APAP toxicity with chronic EtOH ingestion
Herbs and medications	↑ APAP toxicity with impedance of hepatic glucouronidation
Age and genetic factors	↑ APAP toxicity with advancing age and impaired glucouronidation
Nutritional status	↑ APAP toxicity in malnourished patients
Chronic liver disease	↑ APAP toxicity, especially in chronic EtOH abuse



Acetaminophen

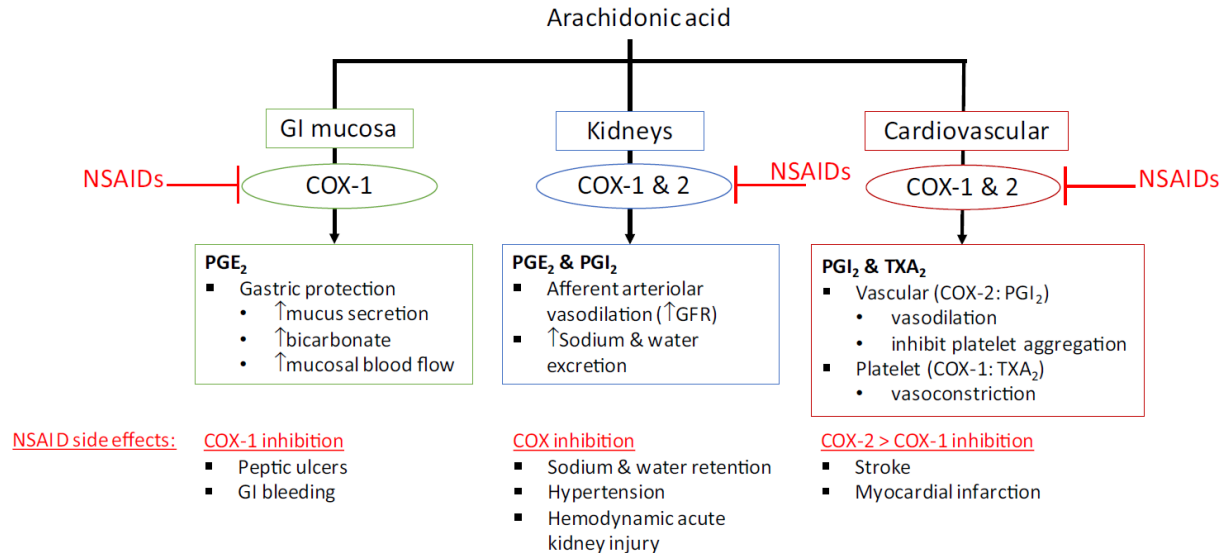
- **Safety considerations**

- Hepatic impairment
 - active liver disease or severe hepatic impairment - contraindicated
 - mild hepatic impairment: use with caution
- Renal impairment
 - severe renal impairment (creatinine clearance ≤ 30 mL/min): extending dosing intervals & reducing total daily dosage may be advisable.
- Drug-drug interactions
 - **Warfarin:** prolonged oral administration at 4000 mg/day has been associated with an elevated INR in patients receiving warfarin.
 - **Alcohol:** chronic alcohol misuse increases the risk of acetaminophen toxicity by inducing CYP2E1, reducing hepatic glutathione (GSH) levels & impairing NAPQI detoxification.



NSAIDs

- First-line for moderate to severe pain.
- Systemic pain relief and anti-inflammatory effects.



NSAIDs



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		Usual dose	Maximum dose
Non-selective NSAIDs	Ibuprofen	200-400 mg q 4-6 hours	3200 mg/day
	Naproxen	250-500 mg q 12 hours	1500 mg/day
	Diclofenac	50 mg 2-3 times daily	150 mg/day
COX-2 selective NSAIDs	Celecoxib	200 mg OD (may increase to BID)	400 mg/day
	Etoricoxib	30 or 60 mg OD	120 mg/day

NSAIDs



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	Side effects	Comments
Non-selective NSAIDs	• Gastrointestinal: COX-1 inhibition causes dyspepsia, (If severe) gastric bleeding, ulceration • Rashes: urticaria, toxic epidermal necrolysis • Acute renal failure • Increase blood pressure • Increase risk of cardiovascular events • Reduce effect of anti-hypertensives	• Contraindicated in active peptic ulceration, bleeding disorder, and heart failure. • Naproxen → lower CV risk.
COX-2 selective NSAIDs	• Similar to other NSAIDs • Less gastrointestinal side effects.	• Contraindicated in active peptic ulceration and cardiovascular disease. • Monitor blood pressure

NSAIDs



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Safety considerations

- **Gastrointestinal risks**
 - GI ulceration/bleeding: Non-selective NSAIDs
 - Preventive measures: Patients at high risk (e.g., elderly, history of ulcers) should be prescribed a proton pump inhibitor (PPI).
- **Cardiovascular risks**
 - Increased risk of CVD events: COX2-selective
 - Non-selective NSAIDs (especially naproxen) are considered safer.
 - Blood pressure: NSAIDs can lead to an increase in blood pressure & may worsen heart failure, especially in chronic use.

NSAIDs



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- **Renal risks**

- Monitoring renal function is important in long-term users, especially in patients with renal disease.

- **Hepatic risks**

- Hepatotoxicity is rare.
- Monitoring liver function tests is advisable for patients on prolonged therapy.

- **Drug interactions**

- Anticoagulants (e.g., warfarin): NSAIDs increase the risk of bleeding, especially in combination with anticoagulants.
 - Monitoring INR is recommended.
- Antihypertensives: NSAIDs may reduce the efficacy of antihypertensive drugs like ACE inhibitors, ARBs, and diuretics.



Topical NSAIDs

- **First-line treatment for mild to moderate knee and hand OA.**
 - especially for older adults & those with comorbidities who are at higher risk of side effects from oral medications.
- Effective for localized and superficial pain.
- Diclofenac gel (1-3%), ketoprofen gel (2.5%)
- Apply to the affected area 3 to 4 times daily
- **Safety considerations**
 - Skin reaction: dry skin, local irritation, rash, or dermatitis at the application site.

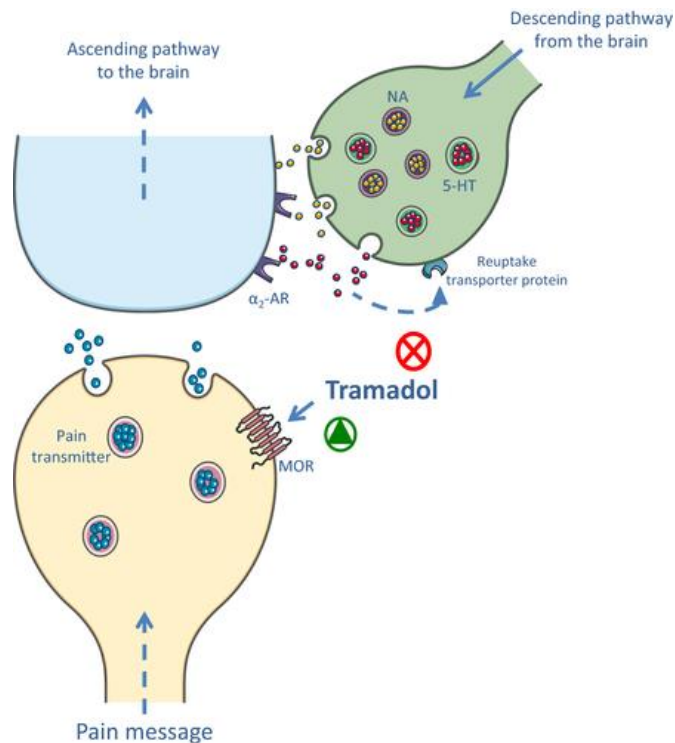


Topical capsaicin

- Commonly used for localized pain (e.g. elbow, knee, ankle, wrist, feet)
- **MOA:** With repeated use, it depletes substance P, reducing the sensation of pain.
- Apply to the affected area 3 - 4 times daily
- It may take several days to a few weeks for pain relief to become noticeable.
- **Safety considerations**
 - Initial burning sensation: This is usually mild and diminishes with continued application.
 - Skin irritation: redness, itching, or swelling
 - Should not be applied to broken/irritated skin.

Tramadol

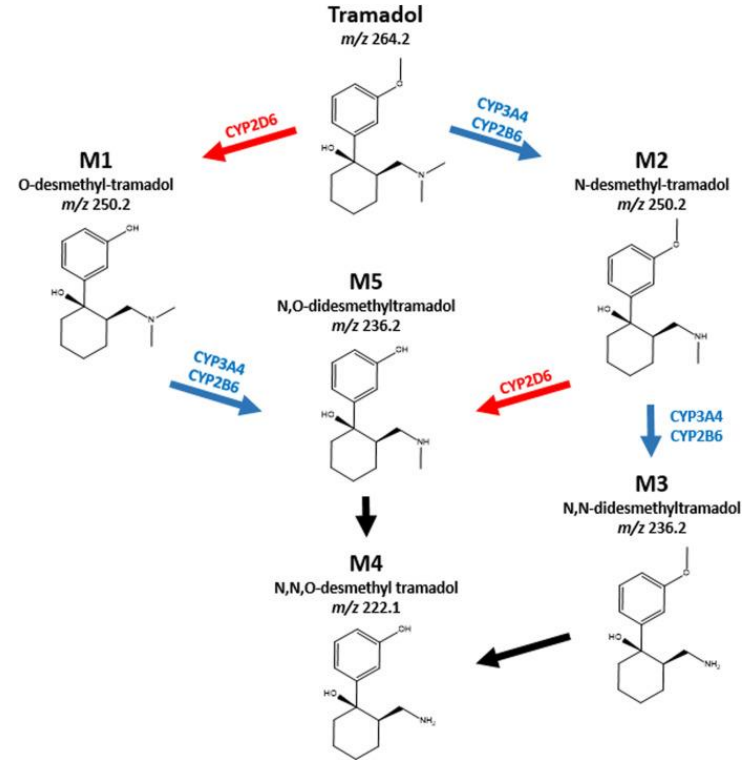
- For moderate to severe pain.
- A **second-line or adjunctive treatment option** when first-line therapies such as acetaminophen and NSAIDs are insufficient for pain control.
- **Dual mechanisms**
 - weak opioid agonist
 - serotonin & norepinephrine reuptake inhibitor
- Dose: 50 mg every 4-6 hours.
- Maximum dose: 400 mg /day.



Tramadol

Pharmacokinetics

- Oral bioavailability: 75%
- Plasma protein binding: 20%
- Metabolism: **CYP2D6**, **CYP3A4** & CYP2B6
 - **O-desmethyl-tramadol (M1): active metabolite**
 - **CYP2D6 poor metabolizer:** reduced M1 → inadequate pain relief
 - **CYP2D6 ultrarapid metabolizer:** increased M1 → increase toxicity risk
- Excretion: mainly renal (30% as unchanged, 60% as metabolites)





Tramadol

- **Drug interactions:**

- **CYP2D6 inhibitors** (e.g., amiodarone, fluoxetine, paroxetine, duloxetine, bupropion) may reduce tramadol's conversion to its active metabolite, decreasing its effectiveness.
- **CYP3A4 inhibitors** (e.g., ketoconazole, erythromycin, clarithromycin, ritonavir) can increase tramadol levels, potentially increasing the risk of side effects.
- **CYP3A4 inducers** (e.g., rifampin, phenytoin, carbamazepine) can decrease tramadol levels, reducing efficacy.

- **Dosing considerations:**

- Elderly patients: start at lower doses & monitor closely (maximum dose: 300 mg/day).
- Renal impairment: reduce dose in patients with a creatinine clearance <30 mL/min.
- Hepatic impairment: dose adjustments may be necessary, as tramadol is metabolized by the liver.

Tramadol



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Safety considerations:

- GI effects: nausea, vomiting, constipation
- CNS effects: dizziness, drowsiness
- Serotonin syndrome: when combined with other serotonergic drugs
 - Serotonin syndrome: agitation, confusion, fever, shivering, sweating, and muscle rigidity.
- Respiratory depression: high dose or in combination with other CNS depressants.
- Drug dependency and abuse
 - Abrupt discontinuation can lead to withdrawal symptoms (e.g., anxiety, insomnia, sweating, tremors).
 - Dose should be gradually tapered to prevent withdrawal.



Duloxetine

- **Serotonin-norepinephrine reuptake inhibitor (SNRI)**
- is approved for chronic musculoskeletal pain, including knee OA and chronic low back pain.
- May be prescribed for patients who
 - have chronic OA pain that is not adequately controlled with acetaminophen or NSAIDs.
 - cannot tolerate NSAIDs or tramadol due to gastrointestinal, cardiovascular, or addiction-related concerns.
- **Dose:**
 - Initial dose: 30 mg once daily. Maintenance dose: after 1-2 weeks, the dose may be increased to 60 mg/day.
 - Maximum dose: 60 mg/day



Duloxetine

- Pain relief typically begins within 1-2 weeks, but it may take up to 4-6 weeks for the full effects to be realized.
- **Drug interactions:**
 - CYP2D6 inhibitors: Duloxetine is metabolized by CYP2D6, and its levels can be affected by CYP2D6 inhibitors.
 - MAO inhibitors: increase risk of hypertensive crisis or serotonin syndrome.
 - Alcohol: increases the risk of liver damage
 - patients should be advised to limit or avoid alcohol consumption.
- **Common side effects:** headache, nausea (usually resolved within days to weeks), drowsiness, fatigue, dry mouth, constipation, diarrhea.
- **Serious side effects:** suicidality, serotonin syndrome, hepatotoxicity



Intraarticular (IA) corticosteroid

- **Anti-inflammatory and pain relief effects.**
- **Indications:**
 - Local inflammation: For patients with OA who experience significant inflammation and pain in specific joints, such as the knee, hip, or shoulder.
 - Alternative to oral medication.
 - Short-term relief: enabling patients to participate in physical therapy or other rehabilitation efforts.
- Onset of action: Pain relief usually begins within 24-48 hours of injection, with maximal effects occurring within 1-2 weeks.



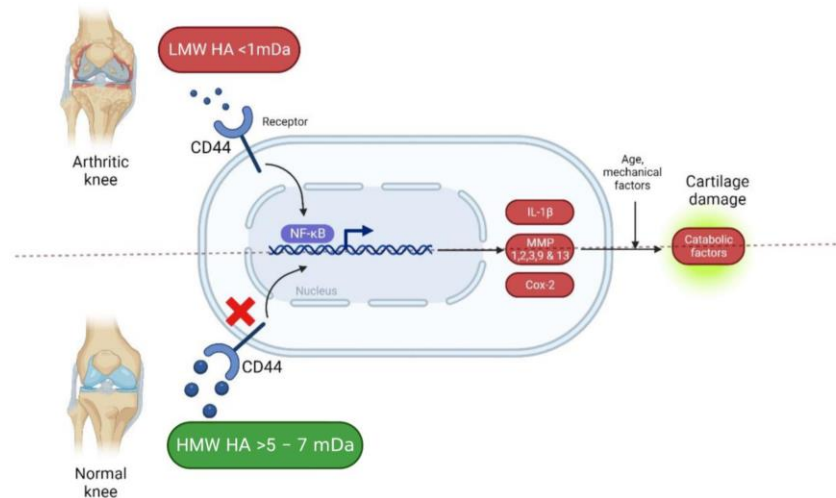
Intraarticular (IA) corticosteroid

- **Post-injection care:**
 - Rest the joint for the first 24 hours.
 - Monitor for signs of infection.
 - Avoid strenuous activities.
- **Safety considerations**
 - Local side effects: pain or swelling, joint infection, cartilage damage (repetitive use)
 - Maximum of 3 to 4 injections per year in a single joint.
 - Risk of osteoporosis: long-term use of systemic corticosteroids is associated with decreased bone density.



IA Hyaluronic acid

- Act as a lubricant and shock absorber in the joints.
- Improve quality & quantity of synovial fluid – reduce pain & improve mobility.
- Commonly used in knee OA.
- Single intraarticular injection → pain relief for 6 months to 1 year



Glucosamine



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- Currently no FDA-approved indication for glucosamine in OA treatment.
- Effectiveness is mixed.
- Commonly used as an oral supplement.
- Precursor for synthesis of glycosaminoglycans – a major component of cartilage.
- Cartilage protection – promote cartilage repair.
- Dose: 500 mgTID or 1,500 mg/day
- Often used for 3-6 months, mild to moderate OA.
- Side effects: Epigastric pain, heartburn, diarrhea, nausea

Chondroitin



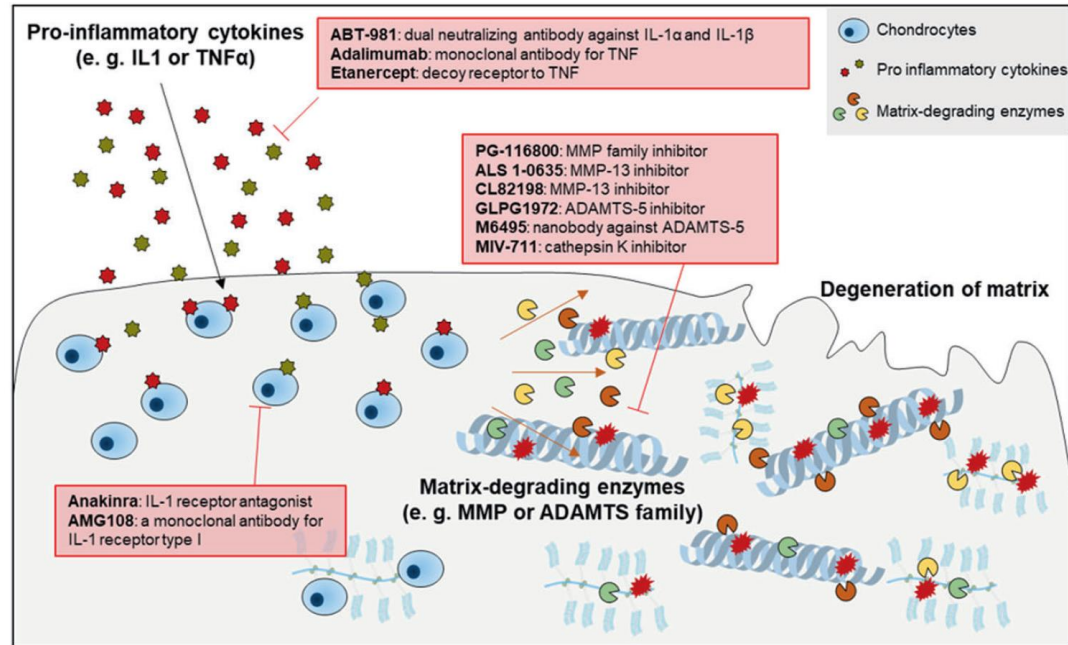
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- Effectiveness is mixed.
- Chondroitin is a major component of cartilage.
- Prevent cartilage breakdown.
 - Chondroitin may reduce the activity of enzymes that destroy cartilage and collagen in the joints.
- May stimulate the body's repair mechanisms to rebuild cartilage.
- Dose: 200-400 mg BID or TID (800-1200 mg/day)
- Side effects: stomach upsets, headache, increased intestinal gas, diarrhea



DMOADs: Disease-Modifying Osteoarthritis Drugs

Blocking pro-inflammatory cytokines and matrix-degrading enzymes





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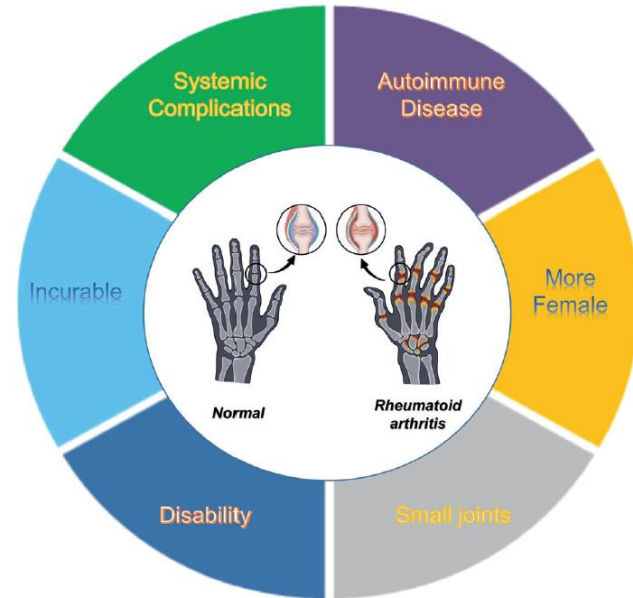
Rheumatoid Arthritis

โรคข้ออักเสบรูมาตอยด์

Rheumatoid Arthritis (RA)

- Definition: An **autoimmune disease** in which the body's immune system begins attacking the lining of its joints, causing **pain, inflammation**, and, eventually, **damage to cartilage and bone**.
- Epidemiology: 1% of the adult population worldwide.
- Unknown etiology.
- **Symptoms:** symmetrical joint pain, joint inflammation (tenderness, swelling, erythema, warmth), stiffness (lasts more than 30 minutes), limited range of motion, low-grade fever, fatigue

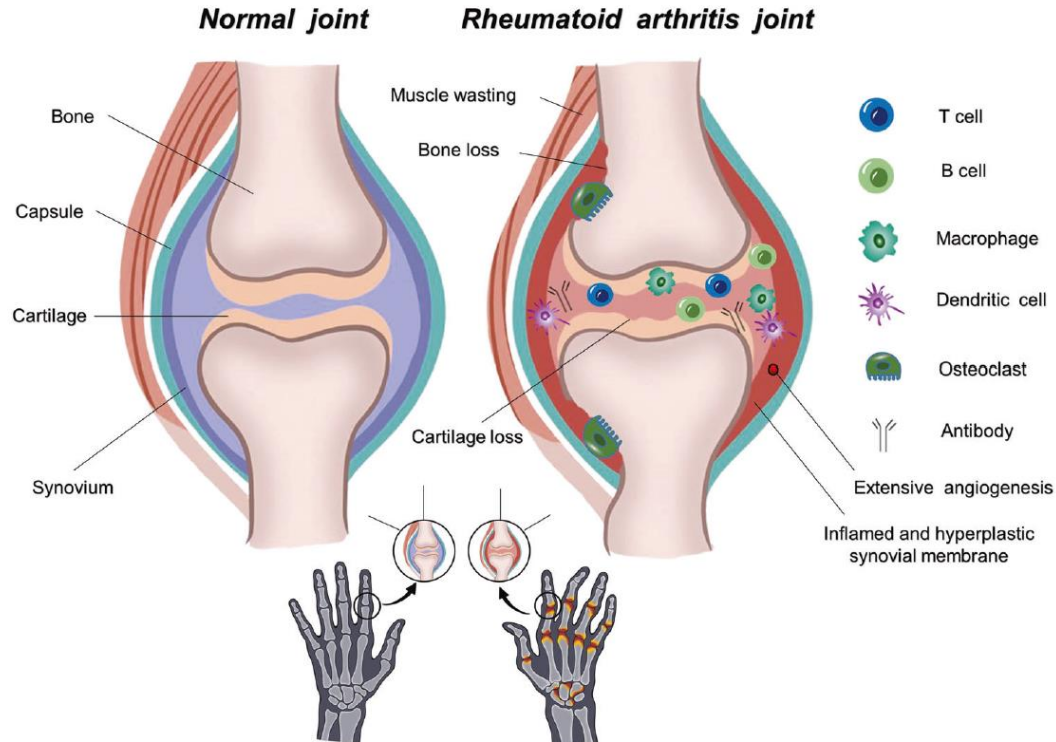
RA Features



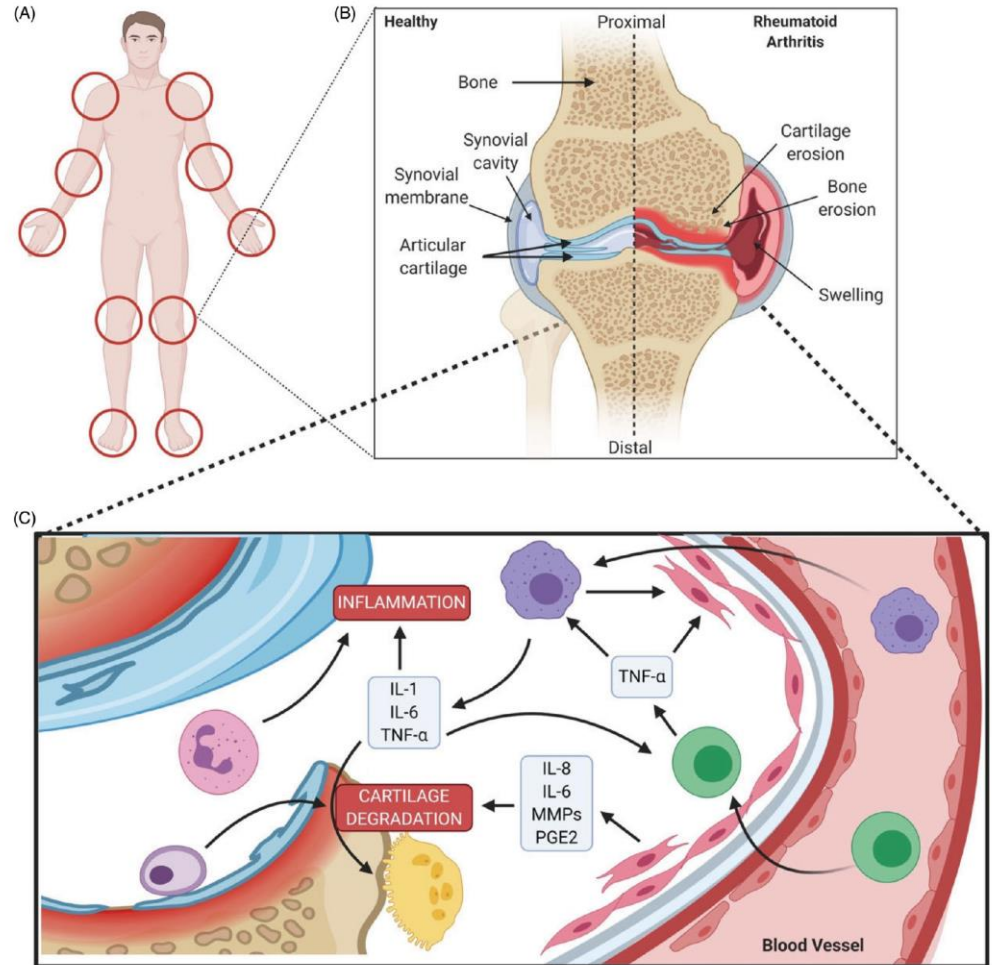
Normal vs RA joint



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Pathophysiology of RA





RA Treatment

- **Goal:** To improve or maintain the patient's quality of life.
 - Reduce pain and inflammation.
 - Prevent joint damage and deformities.
 - Preserve joint function and mobility.
 - Achieve disease remission or low disease activity.
 - Delay or prevent disability and long-term complications.



RA Treatment

Non-pharmacological

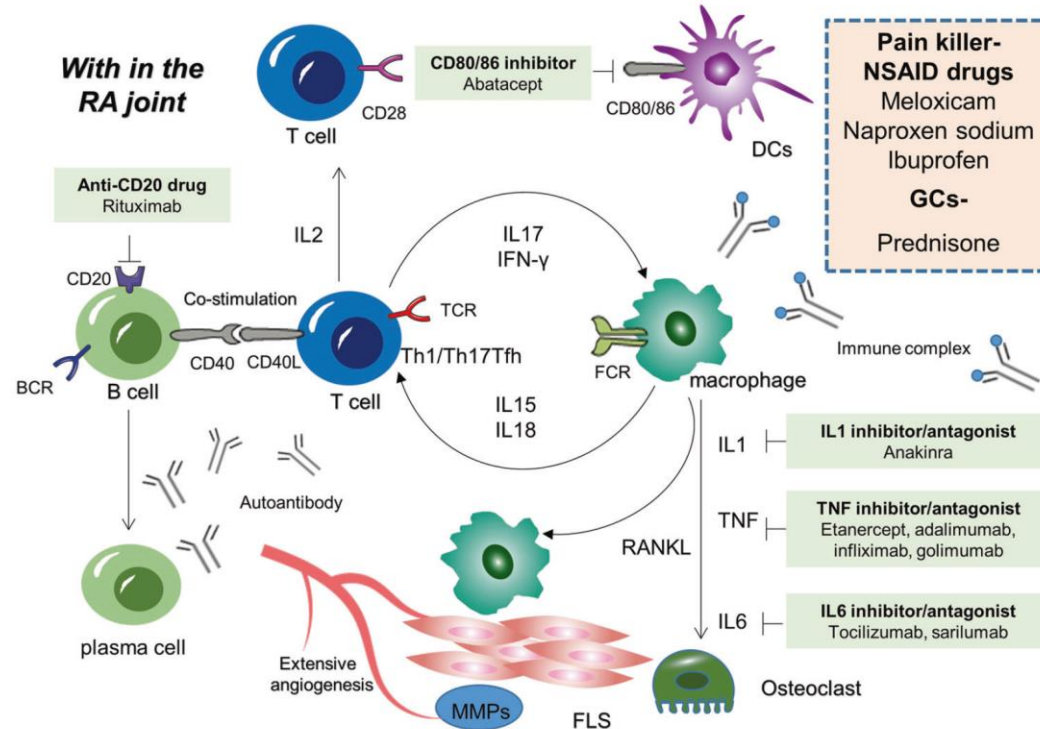
- Patient education
- Physical therapy & Exercise
- Lifestyle modification
- Heat and cold therapy

Pharmacological

- NSAIDs
- Corticosteroids
- DMARDs
 - Conventional synthetic
 - Biologics
 - Targeted synthetic



Pathophysiology & drug targets



Pharmacological Treatment



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- NSAIDs
- Corticosteroids

Anti-inflammation
“short onset of action”

- **Conventional synthetic DMARDs (csDMARDs):**

- Methotrexate (MTX)
- Sulfasalazine
- Leflunomide
- Hydroxychloroquine

- **Biologic DMARDs (bDMARDs):**

- Tumor necrosis factor (TNF) inhibitors: etanercept, adalimumab, infliximab, golimumab, certolizumab
- Interleukin inhibitors (e.g. anakinra, tocilizumab)
- B-cell depleting agents (e.g., rituximab)
- T-cell costimulatory blockers (e.g., abatacept)

- **Targeted synthetic DMARDs (tsDMARDs)**

- Janus kinase (JAK) inhibitors (e.g., tofacitinib, baricitinib, upadacitinib)

DMARDs:
Disease-Modifying
Antirheumatic Drugs
“slow onset of action”



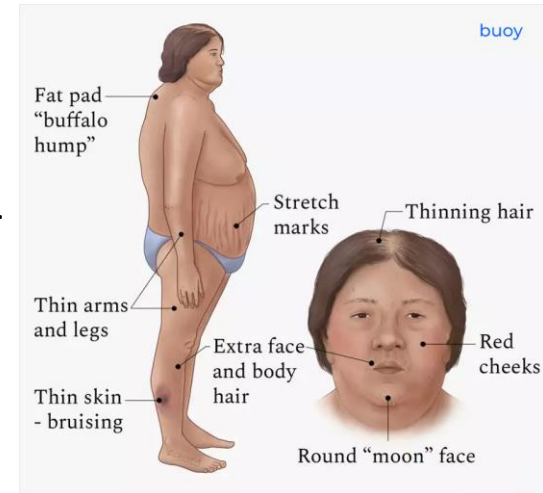
NSAIDs

- Analgesic and anti-inflammatory effects without preventing joint damage.
- Treat mild to moderate pain, swelling, and joint stiffness.
- Non-selective: ibuprofen, naproxen, meloxicam
- COX-2 inhibitor: celecoxib
- Safety considerations:
 - GI, CVD and CKD risk
 - PPI for patients with high GI risk receiving non-selective NSAIDs.



Corticosteroids

- Inhibit inflammation & prevent disease progression.
- Prednisolone
- Provide quick relief from acute inflammation.
- Use as low dose and infrequently as possible.
- Often use while waiting for DMARDs to take effect.
- Not suitable for long-term use.
- Dose: 5-10 mg/day, may increase up to 60 mg/day.
- Side effects (long-term use): osteoporosis, weight gain, hypertension, hyperglycemia, cushing's syndrome.





Corticosteroids

- Safety consideration:
 - Patients with and without osteoporosis risk factors on low-dose prednisone should undergo bone densitometry (DEXA Scan) to assess fracture risk.
 - Bisphosphonates (e.g., alendronate, risedronate, ibandronate) are recommended to prevent and/or treat osteoporosis in addition to adequate calcium and vitamin D supplementation.
- **Intra-articular corticosteroids** (e.g., triamcinolone, methylprednisolone) are effective for controlling a local flare in a joint without changing the overall drug regimen.

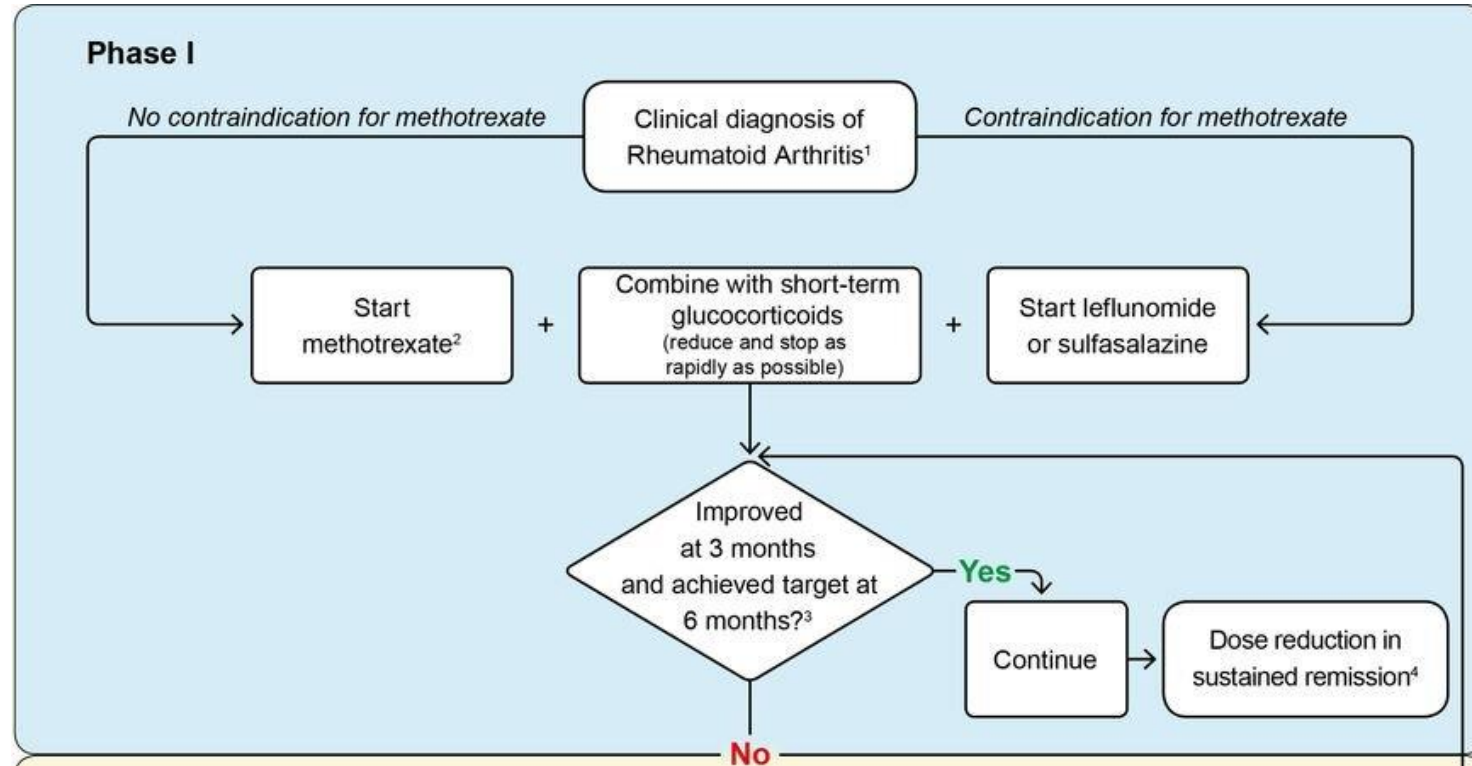
DMARDs

- **Conventional synthetic DMARDs (csDMARDs):**
 - Methotrexate (MTX)
 - Sulfasalazine
 - Leflunomide
 - Hydroxychloroquine
- **Biologic DMARDs (bDMARDs):**
 - Tumor necrosis factor (TNF) inhibitors: etanercept, adalimumab, infliximab, golimumab, certolizumab
 - Interleukin inhibitors (e.g. anakinra, tocilizumab)
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- **Targeted synthetic DMARDs (tsDMARDs)**
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Pharmacological treatment



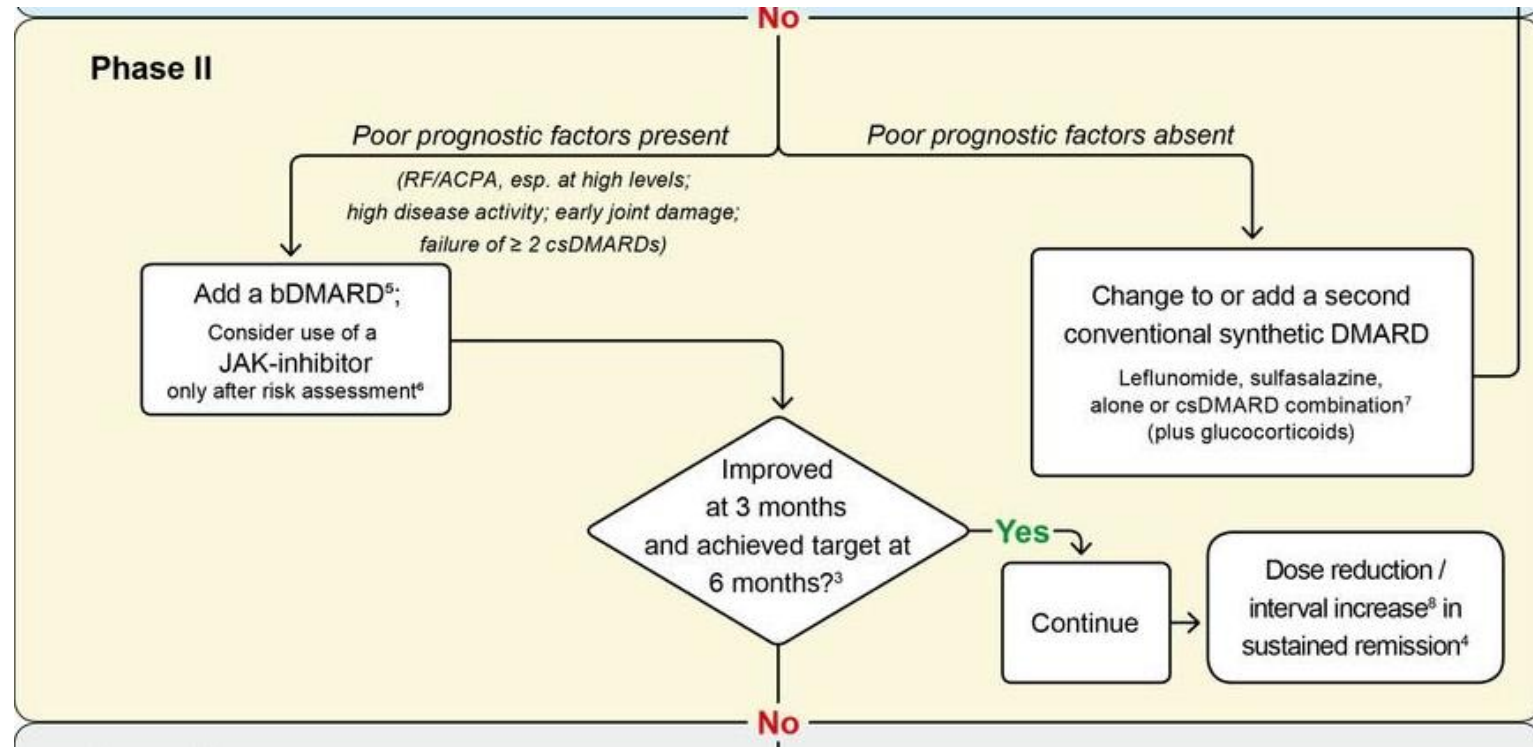
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Pharmacological Treatment



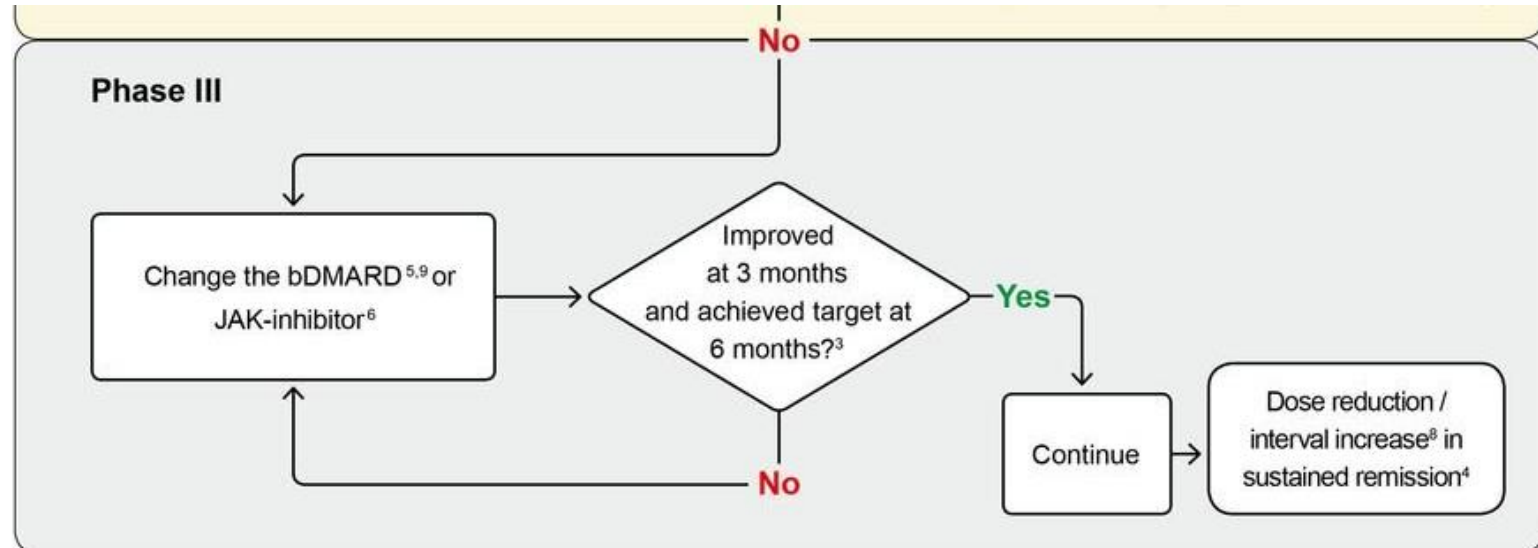
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Pharmacological Treatment



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Conventional synthetic DMARDs



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Methotrexate (MTX)

- **First-line DMARDs**
- Rapid onset, high efficacy, cheap
- MOA: inhibit enzymes responsible for nucleotide synthesis → reduce DNA synthesis → inhibit rapidly dividing cells (e.g., immune cells) → reduce inflammation.
 - dihydrofolate reductase (DHFR),
 - methylenetetrahydrofolate reductase (MTHFR),
 - thymidylate synthase (TYMS),
 - AICAR transformylase (ATIC),
- Dose: 7.5 – 25 mg/week (once a week)
- Onset of action: 4-6 weeks

- [illegible]

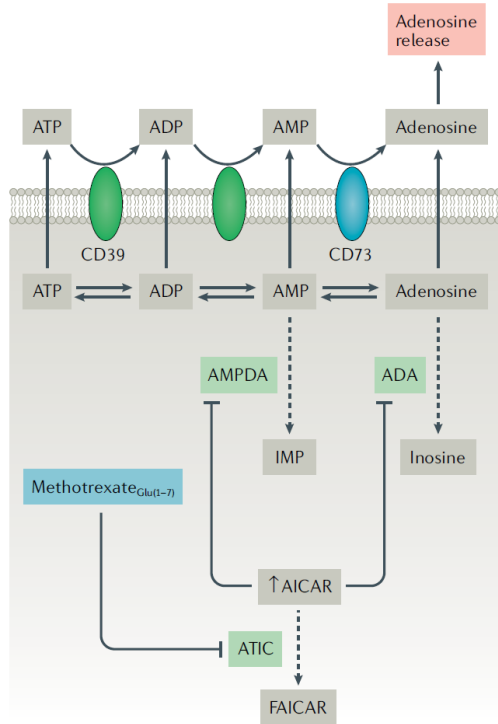
ATIC: AICAR transformylase



Methotrexate (MTX)



- 1) Folate pathway
- 2) Adenosine signaling pathway



Adenosine accumulation



Bind with adenosine receptors on immune cells (e.g., T-cell)



Inhibitory effects on inflammatory cells

- suppress T-cell, B-cell activation.
- inhibit neutrophil adhesion & recruitment.
- inhibit cytokine production. (e.g., TNF- α , IL-6, and IL-8)



Conventional synthetic DMARDs

Methotrexate (MTX)

- Pharmacokinetics
 - Oral bioavailability: 60%
 - Plasma-protein binding: 50%
 - Metabolism: mainly liver
 - Excretion: mainly renal
- Side effects: mucosal ulcers, nausea, vomiting, loss of appetite, hair thinning or hair loss.
- Safety considerations:
 - Folate supplementation (5 - 7 mg weekly) should be considered for all patients to prevent bone marrow suppression.
 - Hepatotoxicity: ↑ LFTs, risk of fibrosis/cirrhosis with long-term use.
 - Teratogenic – contraindicated in pregnancy

Conventional synthetic DMARDs



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Methotrexate (MTX)

- Safety considerations:
 - Monitoring:
 - Baseline: CBC, liver function tests, renal function, chest X-ray (for lung disease risk)
 - Regular: CBC, LFTs, renal function every 1-3 months.
- Drug interactions:
 - NSAIDs, salicylates, TMP, PPIs → increase MTX levels
 - Warfarin → increase MTX levels
 - Chloramphenicol, tetracycline → decrease MTX intestinal absorption → decrease MTX levels
 - Alcohol, leflunomide → increase liver toxicity



Conventional synthetic DMARDs

Sulfasalazine

- Second choice
- Has both anti-inflammatory & immunosuppressive effects.
- Has lower efficacy than MTX.
- Exact mechanism is unclear, proposed MOA include:
 - ↓ T-cell proliferation and cytokine release
 - Inhibition of NF- κ B and prostaglandin synthesis
 - Antioxidant effect (scavenging reactive oxygen species)
- Dose: 500 mg OD or 1-3 g/day
- Onset of action: 1-3 months



Conventional synthetic DMARDs

Sulfasalazine

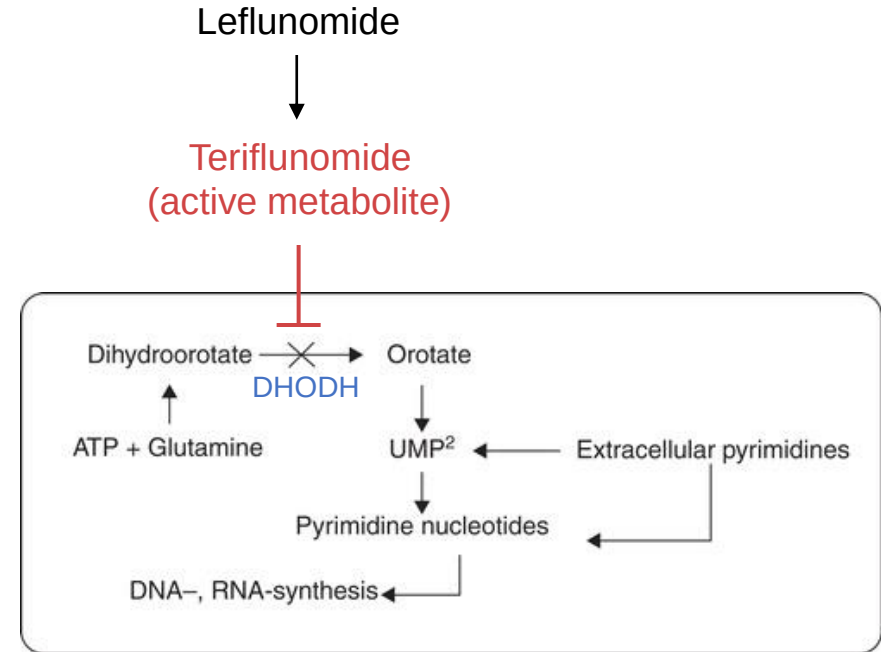
- Side effects: nausea, vomiting, loss of appetite, dyspepsia, rash, headache, discolored urine/skin (orange-colored urine)
 - Low myelosuppression and hepatotoxicity.
- Contraindication: G-6-PD deficiency, sulfa allergy
- Safe in pregnancy and lactation.
- Monitoring
 - CBC, liver function, renal function (baseline, then every 1–3 months)
 - Screen for G6PD deficiency (risk of hemolysis)
- is a preferable agent to MTX in patients with liver disease or in patients who have hepatitis B or C.



Conventional synthetic DMARDs

Leflunomide

- MOA: inhibit dihydroorotate dehydrogenase (DHODH), an enzyme involved in pyrimidine synthesis, thereby reducing the proliferation of activated T-cells.
- Onset of action: 4-6 weeks
- Dose:
 - loading dose: 100 mg OD for 3 days
 - maintenance dose: 10 - 20 mg/day - depending on the patient's tolerance



Conventional synthetic DMARDs



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Leflunomide

- Efficacy is similar to MTX. → alternative choice for patients who failed/intolerant to MTX.
- Side effects: nausea, diarrhea, hypertension, alopecia, elevated liver enzymes, abdominal pain, back pain, low bone marrow suppression
- Teratogenic – contraindicated in pregnancy
- Monitoring
 - Liver function tests
 - CBC
 - Blood pressure

Conventional synthetic DMARDs



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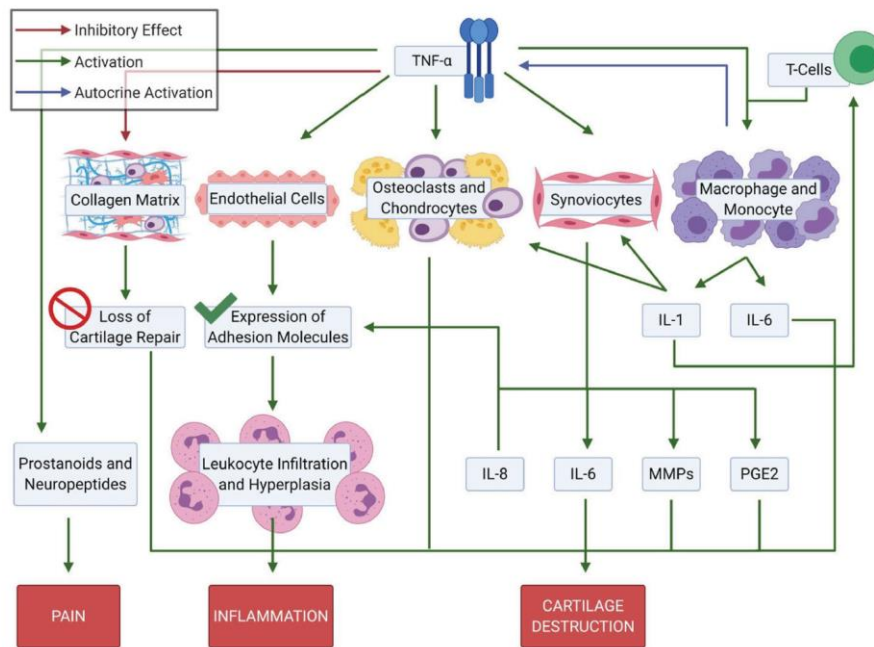
Hydroxychloroquine

- Cheap, moderate efficacy
- MOA: modulates immune response by inhibiting toll-like receptor (TLR) signaling and reducing antigen presentation.
- Dose: 200-400 mg OD or BID
- Onset of action: 2-6 months
- Side effects: nausea, diarrhea, alopecia, visual disturbances (if severe → retinopathy)
- Pregnancy category C (Use in pregnancy is considered acceptable.)

Biologic DMARDs

Biologics targeting specific components of the immune system are typically used when csDMARDs fail.

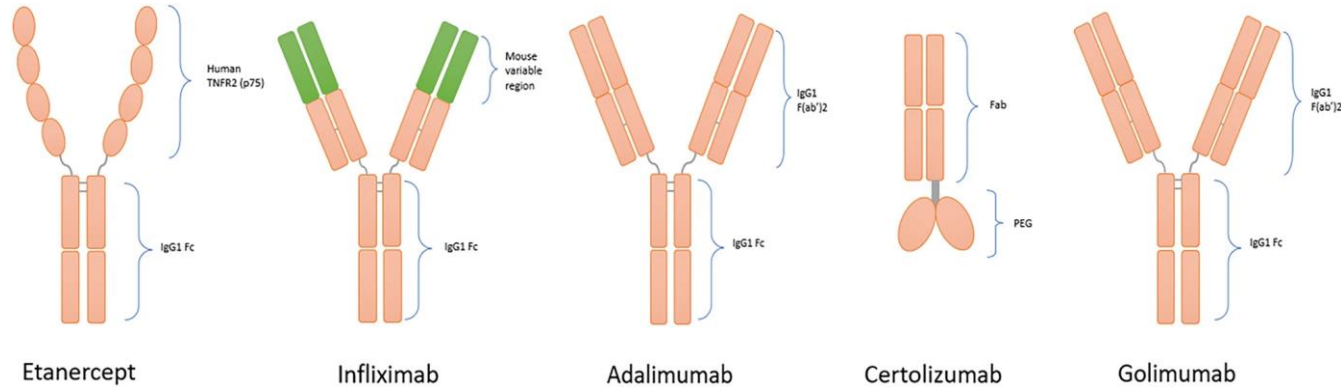
Tumor necrosis factor alpha (TNF α)
- pro-inflammatory cytokine produced by macrophages & lymphocytes



Biologic DMARDs

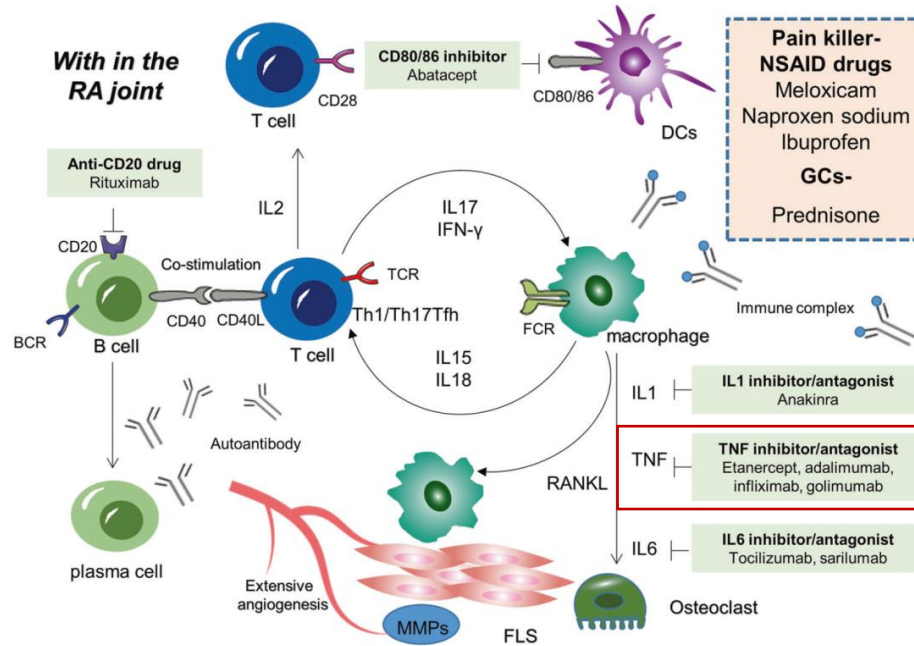
TNF inhibitors - first biological DMARDs to be approved for RA treatment

- MOA: TNF inhibitors block the activity of tumor necrosis factor-alpha (TNF- α), a key cytokine involved in systemic inflammation in RA.



-ximab: chimeric mab
-zumab: humanized mab
-mumab: human mab

TNF inhibitors



TNF inhibitors



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Table 1

List of currently available anti-TNF therapeutics and their approved indications.

CURRENT ANTI-TNF BIOLOGICS				
Drug name & structure	Brand name (Company and first approval date)	Route/Half-life/co-therapy ^a	Approved disease indications	Website
Etanercept Recombinant fusion protein: Human TNFR2:IgG1-Fc	Enbrel [®] (Immunex/now Amgen (US), and Pfizer (UK)) FDA registered Nov.1998	s.c. injection, 2–4 weeks 25 mg vial 50 mg/mL	• Rheumatoid arthritis • Polyarticular juvenile idiopathic arthritis • Psoriatic arthritis • Ankylosing spondylitis • Plaque psoriasis	enbrel.com
Infliximab Humanized (chimeric) IgG1κ mAb	Remicade [®] (Centocor Ortho Biotech Inc. (US) and Janssen-Cilag Pty Ltd. (UK)). FDA registered Aug. 1998	i.v. infusion 4 weeks 100 mg vial	• Rheumatoid arthritis* • Psoriatic arthritis* • Ankylosing spondylitis • Plaque psoriasis • Crohn's disease (moderate/severe) • Pediatric RA & Pediatric Crohn's	remicade.com
Adalimumab Human IgG1κ mAb	Humira [®] (ABBVIE Inc.) FDA registered Dec. 2002.	s.c. injection 2–4 weeks, 40 mg/0.8 mL syringe/vial	• Rheumatoid arthritis* • Psoriatic arthritis* • Plaque psoriasis • Active ankylosing spondylitis • Crohn's disease (moderate/severe) • Juvenile idiopathic arthritis (severe)	humira.com
Golimumab Human IgG1κ mAb	Simponi [®] (Centocor Ortho Biotech Inc.) FDA registered Apr. 2009.	s.c. injection 2 weeks 50 mg injection	• Ulcerative colitis • Rheumatoid arthritis* • Psoriatic arthritis* • Plaque psoriasis • Ulcerative colitis	simponi.com
Certolizumab Pegol Pegylated-Fab' fragment of humanized IgG1κ mAb	Cimzia [®] (UCB Pharma SA) FDA registered Apr. 2008	s.c. injection 200 mg injection	• Rheumatoid arthritis* • Psoriatic arthritis* • Ankylosing spondylitis • Crohn's disease (moderate/severe)	cimzia.com

Biologic DMARDs



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TNF inhibitors

- Rapid onset of action (improvements seen within 2-4 weeks)
- Side effects:
 - Increased risk of infections: tuberculosis, fungal infections
 - Others: Infusion reactions, heart failure exacerbation, and rare reports of demyelinating diseases.
- TB screening is mandatory before starting TNF therapy.
- Hepatitis B should be screened before starting TNF therapy.

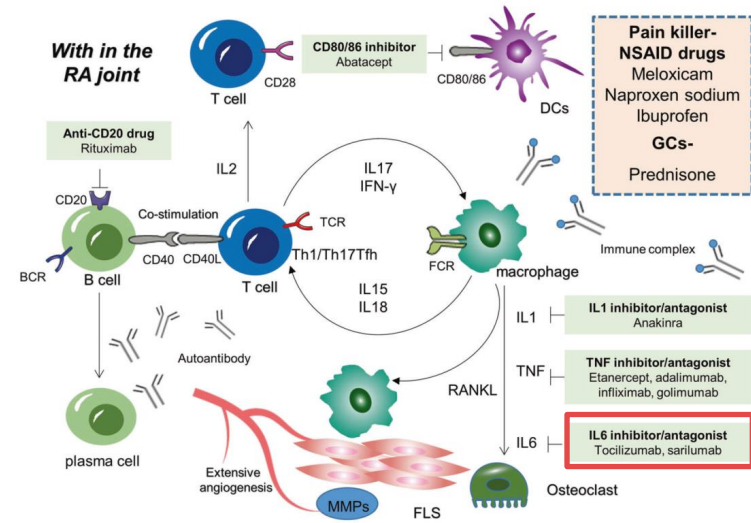
Biologic DMARDs



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Interleukin-6 (IL-6) inhibitors: Tocilizumab

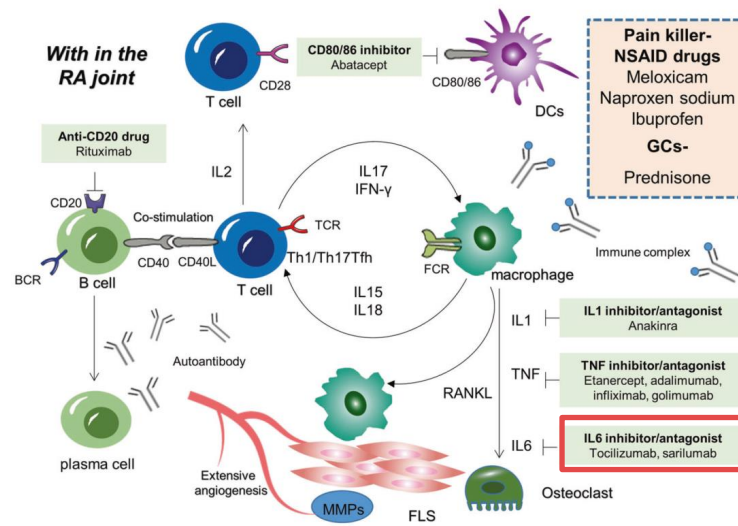
- MOA: binds specifically to both soluble and membrane-bound IL-6 receptors, inhibiting IL-6-mediated signaling.
- Dose:
 - IV: Starting dose 4 mg/kg q 4 weeks, then 8 mg/kg q 4 weeks based on clinical response
 - SC: weight <100 kg 162 mg every other week, weight $\geq$ 100 kg 162 mg weekly
- Onset: 4-8 weeks



Biologic DMARDs

Interleukin-6 (IL-6) inhibitors: Tocilizumab

- Side effects:
 - increased risk of infections
 - neutropenia
 - thrombocytopenia
 - elevations in liver transaminases
 - lipid abnormalities: hyperlipidemia

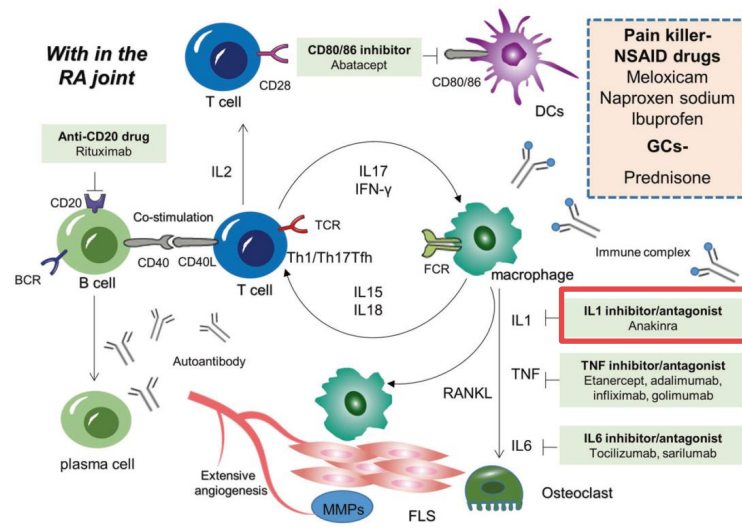




Biologic DMARDs

Interleukin-1 (IL-1) inhibitors: Anakinra

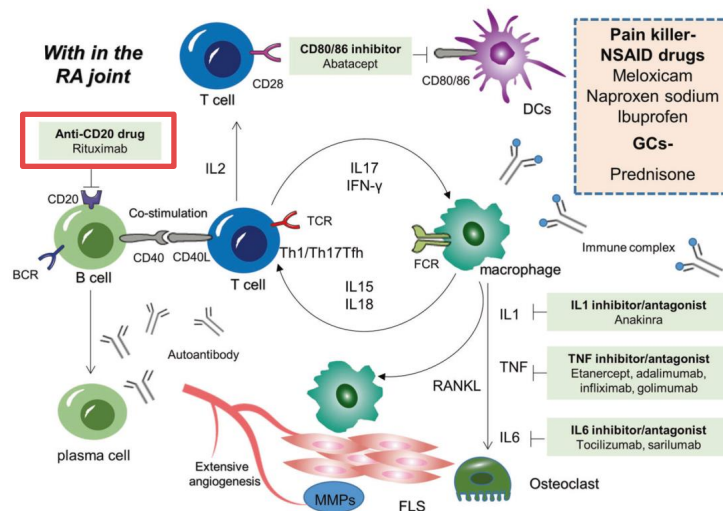
- A recombinant human interleukin-1 receptor antagonist (hu rIL-1Ra)
- MOA: It blocks the binding of IL-1 to its receptor, preventing its pro-inflammatory actions.
- Onset: 2-4 weeks
- Dose: 100 mg SC once daily.
- Side effects:
 - injection site reactions: erythema, itching, discomfort (resolve over 1-2 months)
 - infections: increased risk of infections
 - neutropenia: regular blood count should be tested.



Biologic DMARDs

B-cell Depleting Agents: Rituximab

- MOA: Rituximab targets CD20 on B-cells, leading to B-cell depletion and reduced autoimmune activity.
- Dose: 1000 mg IV, 2 infusions 2 weeks apart, then repeat every 6-12 months.
- Side effects:
 - Infections: Increased risk of infections especially viral reactivation (Hepatitis B).
 - Infusion reaction: fever, chill, hypotension
- Immunizations should be completed before starting therapy. Live virus vaccinations should be avoided.



Ding Q, Hu W, Wang R, et al. Sig Transduct Target Ther 2023;8(68). DOI: 10.1038/s41392-023-01331-9.

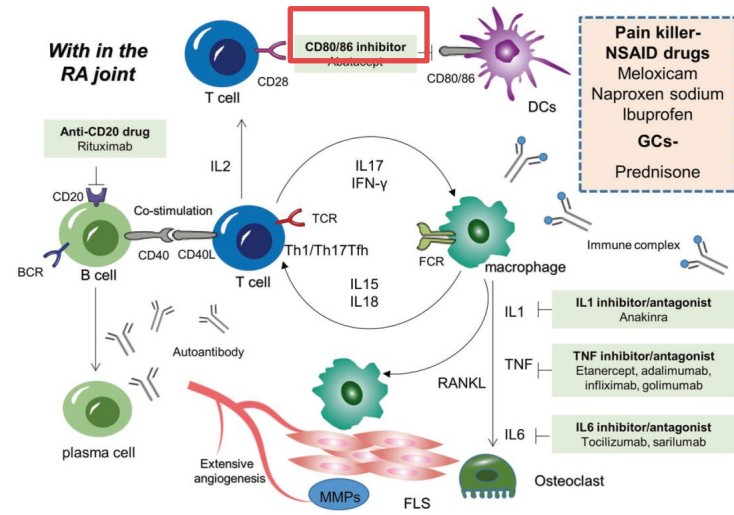
Biologic DMARDs



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T-cell costimulatory blockers: Abatacept

- MOA: binds to CD80 & CD86 on antigen-presenting cells (APCs), preventing their interaction with CD28 on T-cells.
- This blocks the co-stimulatory signal needed for T-cell activation, reducing inflammation & immune response in RA.
- Dose:
 - SC injection: 125 mg once weekly
 - IV injection: loading dose based on body weight (<60 kg: 500 mg, 60-100kg: 750 mg, >100 kg: 1000 mg)
- Responses are typically seen within 3 months.



Ding Q, Hu W, Wang R, et al. Sig Transduct Target Ther 2023;8(68). DOI: 10.1038/s41392-023-01331-9.

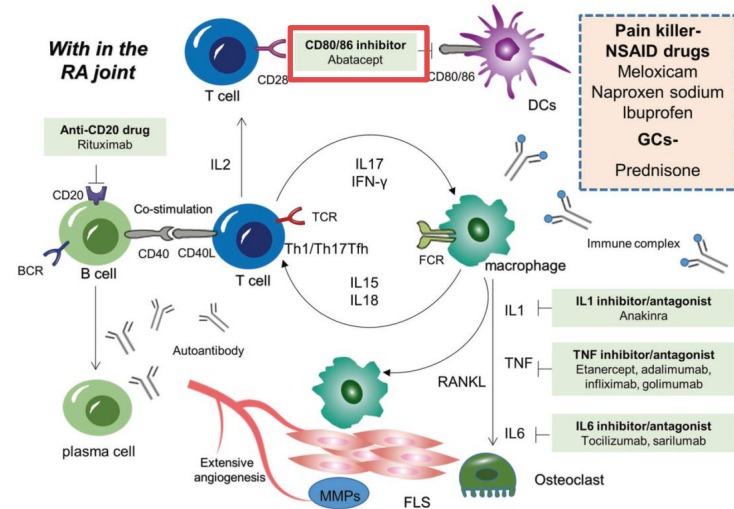
Biologic DMARDs



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T-cell costimulatory blockers: Abatacept

- Side effects:
 - Infections: Increased risk of infections
 - Infusion reaction: more with IV
- Safety consideration:
 - Respiratory infections (pneumonia) were more common in patients with COPD.
 - TB screening is recommended.



Ding Q, Hu W, Wang R, et al. Sig Transduct Target Ther 2023;8(68). DOI: 10.1038/s41392-023-01331-9.

Targeted synthetic DMARDs

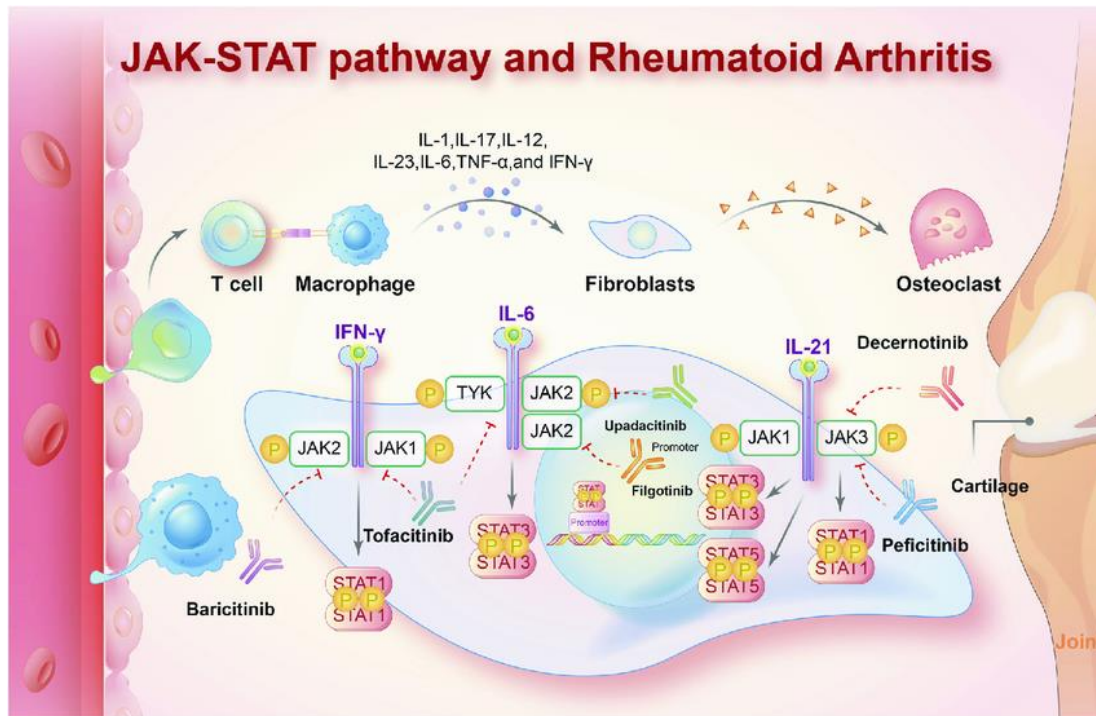


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Janus Kinase (JAK) inhibitors

- MOA: JAK inhibitors block the activity of Janus kinase (JAK) enzymes, which play a key role in the signaling pathways that mediate immune cell activation and inflammation.
- Oral dosage form
- Side effects:
 - Infections: Increased risk of serious infections (herpes zoster)
 - Blood clots: Risk of thrombosis (deep vein thrombosis, pulmonary embolism)
 - Elevated liver enzymes
 - Lipid abnormalities

Janus Kinase (JAK) Inhibitors



Xue C, Yao Q, Gu X, Shi Q, Yuan X, Chu Q, Bao Z, Lu J, Li L. Evolving cognition of the JAK-STAT signaling pathway: autoimmune disorders and cancer. *Signal Transduct Target Ther.* 2023;8(1):204. doi: 10.1038/s41392-023-01468-7.



Janus Kinase (JAK) Inhibitors

JAKi	Manufacturer	JAK selectivity	Dose	Recommended by NICE for RA	FDA approval for RA	EMA approval for RA	Japan approval for RA
Tofacitinib	Pfizer (New York City, USA)	Pan-JAK	5 mg BD 11 mg OD	Yes: severe RA	November 2021	March 2017	March 2013
Baricitinib	Eli Lilly and Company (Indianapolis, Indiana, USA)	JAK1 and JAK2	2 mg OD 4 mg OD	Yes: severe RA	May 2018	February 2017	July 2017
Upadacitinib	AbbVie (Chicago, Illinois)	JAK1	15 mg OD	Yes: severe RA	August 2019	December 2019	January 2020
Filgotinib	Gilead Sciences, Inc. (Foster City, California, USA) and Galapagos Galapagos (Mechelen, Belgium)	JAK1	100 mg OD 200 mg OD	Yes: moderate RA	N/A	September 2020	September 2020
Peficitinib	Astellas Pharma Ltd. (Tokyo, Japan)	JAK3	25 mg OD 50 mg OD 100 mg OD 150 mg OD	N/A	N/A	N/A	March 2019



Comparison: RA & OA

Characteristics	RA	OA
Speed of onset	Rapid (weeks to months)	Slow (years)
Gender prevalence (women:men)	3:1	1-1.5:1
Usual age of onset	30-50 years	Greater than 50
Most common joint affected	Small joints	Large weight-bearing joints
Joint symptoms	Pain, swelling, warmth, stiffness	Pain, bony enlargement
Inflammation	Local and systemic	None or mild
Duration of morning joint stiffness	60 min and longer	Less than 30 min
Joint pattern	Symmetric	Symmetric or asymmetric
ESR	Elevated	Normal
Synovial fluid	Leukocytosis, cloudy	Mild leukocyte
Systemic manifestation	Yes	No



Comparison: RA & OA

RA Pharmacological

- NSAIDs
- Corticosteroids
- DMARDs
 - Conventional synthetic
 - Biologics
 - Targeted synthetic

OA Pharmacological

- | | |
|----------------------|---------------|
| • Acetaminophen | • Glucosamine |
| • NSAIDs | • Chondroitin |
| • Tramadol | • DMOADS |
| • Duloxetine | |
| • IA corticosteroid | |
| • IA hyaluronic acid | |



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THANK YOU
Any questions?